Multiple - correlation

- It shows the relationship b/w at least 3 variables. In other words we can say that, it shows the combined influence of 2 or more independent variables on a single dependendent variable.

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12} \cdot r_{13} \cdot r_{23}}{1 - r_{23}^2}}$$

dependent variable — independent variable

$$R_{2.13} = \sqrt{\frac{r_{21}^2 + r_{23}^2 - 2r_{21} \cdot r_{23} \cdot r_{13}}{1 - r_{13}^2}}$$

$$R_{3.12} = \sqrt{\frac{r_{31}^2 + r_{32}^2 - 2r_{31} \cdot r_{32} \cdot r_{12}}{1 - r_{12}^2}}$$

Note:- Multiple correlation coeff should lie in 0 & 1.

Example Find $R_{1.23}$. Given-that $r_{12} = 0.6$, $r_{13} = 0.7$, $r_{23} = 0.65$

Sol

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12} r_{13} r_{23}}{1 - r_{23}^2}}$$

$$= \sqrt{\frac{(0.6)^2 + (0.7)^2 - 2(0.6)(0.7)(0.65)}{1 - (0.65)^2}}$$

$$= \sqrt{\frac{0.36 + 0.49 - 0.546}{0.5775}}$$

$$= \sqrt{0.5264069264}$$

$$= 0.7255390592$$

Partial Correlation :-

It is a simple correlation b/w two variables after eliminating the influence of the third variable.

In simple words, we can say that it shows the relationship b/w two variables at a time taking other variables as constant.

$$R_{12.3} = \frac{r_{12} - r_{13} \cdot r_{23}}{\sqrt{1-r_{13}^2}\,\sqrt{1-r_{23}^2}}$$

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$$R_{13.2} = \frac{r_{13} - r_{12} \cdot r_{32}}{\sqrt{1-r_{12}^2}\,\sqrt{1-r_{23}^2}}$$

$$R_{23.1} = \frac{r_{23} - r_{21} \cdot r_{31}}{\sqrt{1-r_{21}^2}\,\sqrt{1-r_{31}^2}}$$

Note:- The value of partial correlation coeff should lie b/w -1 to 1.

Q Given that $r_{12} = 0.7$, $r_{13} = 0.61$ and $r_{23} = 0.4$. find $r_{12.3}$

Sol $$r_{12.3} = \frac{r_{12} - r_{13} \cdot r_{23}}{\sqrt{1-r_{13}^2}\,\sqrt{1-r_{23}^2}}$$

$$= \frac{0.7 - (0.61)(0.4)}{\sqrt{1-(0.61)^2}\,\sqrt{1-(0.4)^2}}$$

$$= 0.629$$

Relationship b/w simple, partial and multiple correlation coeff :-

① $1 - R_{1.23}^2 = (1 - r_{12}^2)(1 - r_{13.2}^2)$

② $1 - R_{2.13}^2 = (1 - r_{21}^2)(1 - r_{23.1}^2)$

③ $1 - R_{3.12}^2 = (1 - r_{31}^2)(1 - r_{32.1}^2)$

$$r_{13} = \frac{N\sum X_1X_3 - \sum X_1 \sum X_3}{\sqrt{\{N\sum X_1^2 - (\sum X_1)^2\} - \{N\sum X_3^2 - (\sum X_3)^2\}}}$$

$$= \frac{(10 \times 6137) - (617 \times 98)}{\sqrt{\{(10 \times 38753) - (617 \times 617)\} - \{(10 \times 990) - (98 \times 98)\}}}$$

$$= \frac{904}{\sqrt{684 \times 296}} = \frac{904}{1423.00} = 0.64$$

$\Rightarrow r_{13} = 0.64$

$$\& \; r_{23} = \frac{N\sum X_2X_3 - \sum X_2 \sum X_3}{\sqrt{\{N\sum X_2^2 - (\sum X_2)^2\} - \{N\sum X_3^2 - (\sum X_3)^2\}}}$$

$$= \frac{(10 \times 5178) - (521 \times 98)}{\sqrt{\{10 \times 27423) - (521 \times 521)\} - \{(10 \times 990) - (98 \times 98)\}}}$$

$$= \frac{722}{\sqrt{2789 \times 296}} = \frac{722}{908.59} = 0.79$$

Now

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

$$= \sqrt{\frac{(0.80)^2 + (0.64)^2 - 2(0.80)(0.64)(0.79)}{1 - (0.79)^2}}$$

$$= \sqrt{\frac{0.64 + 0.41 - 0.81}{1 - 0.62}} = \sqrt{\frac{0.24}{0.38}} = \sqrt{0.63}$$

$R_{1.23} = 0.7937 = 0.79$

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{13}^2}} = \sqrt{\frac{0.8^2 + 0.79^2 - 2(0.8)(0.64)(0.79)}{1 - 0.64^2}}$$

$$= \sqrt{\frac{0.64 + 0.62 - 0.81}{1 - 0.49}} = \sqrt{\frac{0.45}{0.51}} = \sqrt{0.88} = 0.94$$

$R_{2.13} = 0.94$

Q. From the following data, obtain $R_{1.23}$, $R_{2.13}$, $R_{3.12}$

X_1	2	5	7	11
X_2	3	6	10	12
X_3	1	3	6	10

Sol

X_1	X_2	X_3	X_1^2	X_2^2	X_3^2	X_1X_2	X_1X_3	X_2X_3
2	3	1	4	9	1	6	2	3
5	6	3	25	36	9	30	15	18
7	10	6	49	100	36	70	42	60
11	12	10	121	144	100	132	110	120
25	31	20	199	289	146	238	169	201

$$r_{12} = \frac{N\Sigma X_1X_2 - \Sigma X_1 \Sigma X_2}{\sqrt{\{N\Sigma X_1^2 - (\Sigma X_1)^2\}\{N\Sigma X_2^2 - (\Sigma X_2)^2\}}} = \frac{4 \cdot 238 - 25 \cdot 31}{\sqrt{\{4 \cdot 199 - 25 \cdot 25\}\{4 \cdot 289 - 31 \cdot 31\}}}$$

$$= \frac{177}{\sqrt{\{171\}\{195\}}} = \frac{177}{182.61} = 0.97$$

$\Rightarrow r_{12} = 0.97$

$$r_{13} = \frac{N\Sigma X_1X_3 - \Sigma X_1 \Sigma X_3}{\sqrt{\{N\Sigma X_1^2 - (\Sigma X_1)^2\} - \{N\Sigma X_3^2 - (\Sigma X_3)^2\}}} = \frac{(4)(169) - (25)(20)}{\sqrt{(4(199) - 25(25))\,(4(146) - 20(20))}}$$

$$= \frac{176}{\sqrt{171 \cdot 184}} = \frac{176}{177.38} = 0.99$$

$$r_{23} = \frac{N\Sigma X_2X_3 - \Sigma X_2 \Sigma X_3}{\sqrt{\{N\Sigma X_2^2 - (\Sigma X_2)^2\} - \{N\Sigma X_3^2 - (\Sigma X_3)^2\}}} = \frac{4 \cdot 201 - 31 \cdot 20}{\sqrt{(4 \cdot 289 - 31 \cdot 31) - (4 \cdot 146 - 20 \cdot 20)}}$$

$$= \frac{184}{\sqrt{195 \cdot 184}} = \frac{184}{189.42} = 0.97$$

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}} = \sqrt{\frac{0.97^2 + 0.99^2 - 2\times 0.97\times 0.99\times 0.9}{1 - 0.97^2}}$$

$$= \sqrt{\frac{0.058}{0.059}} = \sqrt{0.98} = 0.99$$

$R_{2.13} = 0.97$, $R_{3.12} = 0.99$

Q The following data is given

X_1	60	68	50	66	60	55	72	60	62	51
X_2	42	56	45	64	50	55	57	48	56	42
X_3	74	71	78	80	72	62	70	70	76	65

Obtain $R_{1.23}$, $R_{2.13}$, $R_{3.12}$

Sol $r_{12} = 0.69$, $r_{13} = 0.22$, $r_{23} = 0.23$

$R_{1.23} = 0.69$, $R_{2.13} = 0.69$, $R_{3.12} = 0.25$